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### Centralized ledger for airline maintenance control

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#### Abstract

Processing a vehicle maintenance transaction is provided. The method comprises receiving the transaction and entering the transaction into a blockchain ledger. In response to the transaction, an enhanced smart contract in the blockchain ledger executes. The enhanced smart contract specifies a number of compliance requirements to complete the transaction and automates a workflow for the transaction according to the compliance requirements. A number of notifications regarding the transaction are generated to a number of subscribers to the blockchain ledger. Confirmation of fulfillment of the compliance requirements is received from the subscribers to complete the transaction. The enhanced smart contract, verifies fulfillment of the compliance requirements, and completion of the transaction is recorded in the blockchain ledger.

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## Background/Summary

### BACKGROUND INFORMATION

#### Technical Field

(1) The present disclosure relates generally to vehicle maintenance, and more specifically to maintaining control over changes to maintenance records.

#### Background

(2) The maintenance activities associated with aircraft maintenance are highly regulated. There are multiple touch points, or sources of record, for recording the maintenance activities, but no single point of truth, which complicates the ability to keep maintenance records consistent. Without a single source of truth, obtaining the complete maintenance historical record is demanding.

#### SUMMARY

(3) An illustrative embodiment provides a computer-implemented method of processing a vehicle maintenance transaction. The method comprises receiving the transaction and entering the transaction into a blockchain ledger. In response to the transaction, an enhanced smart contract in the blockchain ledger executes. The enhanced smart contract specifies a number of compliance requirements to complete the transaction and automates a workflow for the transaction according to the compliance requirements. A number of notifications regarding the transaction are generated to a number of subscribers to the blockchain ledger. Confirmation of fulfillment of the compliance requirements is received from the subscribers to complete the transaction. The enhanced smart contract, verifies fulfillment of the compliance requirements, and completion of the transaction is recorded in the blockchain ledger.

(4) Another illustrative embodiment provides a system for processing a vehicle maintenance transaction. The system comprises a storage device that stores program instructions and one or

more processors operably connected to the storage device and configured to execute the program instructions to cause the system to: receive the transaction; enter the transaction into a blockchain ledger; in response to the transaction, execute an enhanced smart contract in the blockchain ledger, wherein the enhanced smart contract specifies a number of compliance requirements to complete the transaction and automates a workflow for the transaction according to the compliance requirements; generate a number of notifications regarding the transaction to a number of subscribers to the blockchain ledger; receive, from the subscribers, confirmation of fulfillment of the compliance requirements to complete the transaction; verify, by the enhanced smart contract, fulfillment of the compliance requirements; and upon completion of the transaction, record the completion in the blockchain ledger.

(5) Another illustrative embodiment provides a computer program product for processing a vehicle maintenance transaction. The computer program product comprises a computer-readable storage medium having program instructions embodied thereon to perform the steps of: receiving the transaction; entering the transaction into a blockchain ledger; in response to the transaction, executing an enhanced smart contract in the blockchain ledger, wherein the enhanced smart contract specifies a number of compliance requirements to complete the transaction and automates a workflow for the transaction according to the compliance requirements; generating a number of notifications regarding the transaction to a number of subscribers to the blockchain ledger; receiving, from the subscribers, confirmation of fulfillment of the compliance requirements to complete the transaction; verifying, by the enhanced smart contract, fulfillment of the compliance requirements; and upon completion of the transaction, recording the completion in the blockchain ledger.

(6) The features and functions can be achieved independently in various embodiments of the present disclosure or may be combined in yet other embodiments in which further details can be seen with reference to the following description and drawings.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) The novel features believed characteristic of the illustrative embodiments are set forth in the appended claims. The illustrative embodiments, however, as well as a preferred mode of use, further objectives and features thereof, will best be understood by reference to the following detailed description of an illustrative embodiment of the present disclosure when read in conjunction with the accompanying drawings, wherein:

(2) FIG. 1 depicts a block diagram of a maintenance record system in accordance with an illustrative embodiment;

(3) FIG. 2 depicts a diagram of point-to-point interfaces in accordance with the prior art;

(4) FIG. 3 depicts a diagram of a centralized ledger maintenance system in accordance with an illustrative embodiment;

(5) FIG. 4 depicts a diagram of an architecture for a centralized ledger maintenance system in accordance with an illustrative embodiment;

(6) FIG. 5 illustrates the operation of the centralized ledger through the use of enhanced smart contracts in accordance with an illustrative embodiment;

(7) FIG. 6 illustrates examples of types of entries into the centralized ledger in accordance with an illustrative embodiment;

(8) FIGS. 7A-7C illustrate an example process flow for a work order transaction based on the types of entries shown in FIG. 6;

(9) FIG. 8 depicts a flowchart of a process for processing a vehicle maintenance transaction in accordance with an illustrative embodiment; and

(10) FIG. 9 is an illustration of a block diagram of a data processing system in accordance with an illustrative embodiment.

#### DETAILED DESCRIPTION

(11) The illustrative embodiments recognize and take into account that there are multiple touch points, or sources of record, for recording the maintenance activities for vehicles such as aircraft. No single point of truth exists for keeping maintenance records consistent.

(12) The illustrative embodiments also recognize and take into account that multi-partner integrations are expensive to develop and maintain and are error prone. Different integration patterns with distinct data models and business processes prohibit share compliance.

(13) The illustrative embodiments provides a centralized ledger based on distributed ledger technology. The centralized ledger provides all participants access to an immutable record of maintenance actions. The central ledger becomes the single point of truth for all maintenance activity. The central ledger ensures no participant can alter a single transaction after it has been recorded. A new transaction is required to “reverse” or correct an error. A ledger policy defines the authority of entities to record specified transaction entries. These central ledger characteristics ensure that regulatory requirements are followed.

(14) Adding a smart contract to the ledger chain allows automatic trigger of pre-defined business actions. For example, a work order transaction is entered to initiate aircraft maintenance service. The smart contract then triggers a request to have supplemental task cards attached to the work order chain. Another smart contract may simply notify registered subscribers based on additions to the transaction chain.

(15) The illustrative embodiments provide a single view of truth with transparency among business partners. The immutable centralized ledger captures partner compliance actions and allows automated compliance validation and execution processes. The centralized ledger becomes the single point of integration among business partners and eliminates the need to maintain point-to-point partner integrations. New business partners can be added to the centralized ledger at any time. Compliance contracts can be supported as enhanced smart contracts used as ledger entries. Enhanced smart contracts meld typical smart contracts with standardized business process models such as BPMN (Business Process Modeling Notation).

(16) With reference now to FIG. 1, a block diagram of a maintenance record system is depicted in accordance with an illustrative embodiment. Maintenance record system **100** comprises a centralized ledger **102**, which might take the form of a distributed ledger. Centralized ledger **102** comprises a blockchain **104**. Blockchain **104** is able to record transactions across multiple computers or nodes in a network, each of which maintains a copy of the blockchain. Once a transaction is recorded in blockchain **104** it is immutable. All transactions are visible to any party with access to the network.

(17) Centralized ledger **102** records maintenance transaction **106** in blockchain **104**. An enhanced smart contract **110** associated with maintenance transaction **106** is also stored in blockchain **104**. Enhanced smart contract **110** is a self-executing contract with a number of compliance requirements (rules) for completing the maintenance transaction **106** written into its code. Enhanced smart contract **110** melds typical smart contract capabilities with standardized business process models such as BPMN (Business Process Modeling Notation). BPMN is a graphical representation of business processes in a business process diagram (BPD) similar to activity diagrams in unified modelling language (UML). Enhanced smart contract **110** automatically executes when predefined conditions are met.

(18) Compliance requirements **112** might include required tasks **114** for fulfilling maintenance transaction **106**, required knowledge **116** associated with the required tasks **114**, and scheduling **118** of the required tasks **114** for completion. Enhanced smart contract **110** can retrieve the compliance requirements **112** from a number of certified/authorized data sources **134** in the network with the centralized ledger **102**. Using the centralized ledger **102** it is possible for various

certified/authorized agents among subscribers **122** to complete compliance requirements **112**. Furthermore, a subscriber **124** might have view-only access wherein the subscriber receives results of completed requirements or when a ledger entry is made.

(19) Centralized ledger **102** might have a number of subscribers **122** (see FIG. 3). Each subscriber **124** has a respective area of responsibility **126** related to fulfillment of the maintenance transaction **106** and access permission **128** that defined the subscribers rights of access to records within the blockchain **104** in accordance with access permission policies **120** of the enhanced smart contract **110**.

(20) Subscribers **122** have a single point of access to the centralized ledger **102** through a ledger application programming interface (API) **130** (see FIG. 4). By acting as the single point of connection for all subscribers, centralized ledger **102** eliminates the need for multiple point-to-point interfaces between subscribers **122**. Centralized ledger **102** might generate a resource planning twin **132** of the subscribers **122** (see FIG. 5).

(21) Enhanced smart contract **110** generates transaction notifications **136** to subscribers **122** as different phases of maintenance transaction **106** are completed (see FIGS. 6 and 7). These transaction notifications **136** might be sent to subscribers according to their respective area of responsibility **126** in relation to the maintenance transaction **106** as well as according to their access permissions. Conversely, enhanced smart contract **110** also receives transaction fulfillment notifications **138** in turn from the different subscribers **122** as different phases of the maintenance transaction **106** are completed.

(22) When the maintenance transaction **106** has been completed after satisfaction of all compliance requirements **112**, centralized ledger **102** records a completion record **108** in blockchain **104**. Completion record **108** cannot be changed once entered in the blockchain. The only way to correct or negate the complete record **108** is to create a new transaction.

(23) Maintenance record system **100** can be implemented in software, hardware, firmware, or a combination thereof. When software is used, the operations performed by maintenance record system **100** can be implemented in program code configured to run on hardware, such as a processor unit. When firmware is used, the operations performed by maintenance record system **100** can be implemented in program code and data and stored in persistent memory to run on a processor unit. When hardware is employed, the hardware can include circuits that operate to perform the operations in maintenance record system **100**.

(24) In the illustrative examples, the hardware can take a form selected from at least one of a circuit system, an integrated circuit, an application specific integrated circuit (ASIC), a programmable logic device, or some other suitable type of hardware configured to perform a number of operations. With a programmable logic device, the device can be configured to perform the number of operations. The device can be reconfigured at a later time or can be permanently configured to perform the number of operations. Programmable logic devices include, for example, a programmable logic array, a programmable array logic, a field programmable logic array, a field programmable gate array, and other suitable hardware devices. Additionally, the processes can be implemented in organic components integrated with inorganic components and can be comprised entirely of organic components excluding a human being. For example, the processes can be implemented as circuits in organic semiconductors.

(25) Computer system **150** is a physical hardware system and includes one or more data processing systems. When more than one data processing system is present in computer system **150**, those data processing systems are in communication with each other using a communications medium. The communications medium can be a network. The data processing systems can be selected from at least one of a computer, a server computer, a tablet computer, or some other suitable data processing system.

(26) As depicted, computer system **150** includes a number of processor units **152** that are capable of executing program code **154** implementing processes in the illustrative examples. As used herein

a processor unit in the number of processor units **152** is a hardware device and is comprised of hardware circuits such as those on an integrated circuit that respond and process instructions and program code that operate a computer. When a number of processor units **152** execute program code **154** for a process, the number of processor units **152** is one or more processor units that can be on the same computer or on different computers. In other words, the process can be distributed between processor units on the same or different computers in a computer system. Further, the number of processor units **152** can be of the same type or different type of processor units. For example, a number of processor units can be selected from at least one of a single core processor, a dual-core processor, a multi-processor core, a general-purpose central processing unit (CPU), a graphics processing unit (GPU), a digital signal processor (DSP), or some other type of processor unit.

(27) FIG. 2 depicts a diagram of point-to-point interfaces in accordance with the prior art. The example in FIG. 2 is related to aircraft maintenance. As shown, each airline **204**, **206** and maintenance and repair organizations (MRO) **208** has direct, point-to-point interfaces with each other as well as to a number of maintenance applications **202** related to aircraft maintenance. In addition, the maintenance applications **202** have point-to-point interfaces with each other.

(28) Point-to-point interfaces are expensive to develop and maintain, error prone, and fragile and not easily upgradeable. These shortcomings can result in delays in maintenance planning and execution and present a fractured version of truth where only a subgroup of connected systems receives updates. Another barrier of point-to-point interfaces is the complexity of enhancing functionality (i.e., exchange of additional data). Integration errors can also lead to high costs for airline maintenance operations. Failure of these integration channels can lead to service outages and possible grounding of airplanes, e.g., aircraft on ground (AOG) events, when critical data is not available for servicing an aircraft.

(29) Deriving maintenance information from point-to-point communication among maintenance portfolio products yields no single view of truth for all parties/platforms, which can lead to data consistency errors. This lack of a single view of truth contributes to services outages and grounding of aircraft.

(30) The respective maintenance applications **202** often employ different respective integration patterns. Some of the maintenance applications **202** have multiple touch points with an airline's back office maintenance planning and execution system, all using distinct data models.

(31) In addition, point-to-point interfaces can require lengthy implementation, requiring from 12 months up to several years based on real projects data and often involves semi-manual data loading.

(32) FIG. 3 depicts a diagram of a centralized ledger maintenance system in accordance with an illustrative embodiment. In the present example, centralized ledger maintenance system **300** relates to aircraft maintenance and is an example implementation of maintenance record system **100** in FIG. 1.

(33) The centralized ledger **302** provides a standardized interface via a shared connection point. Centralized ledger **302** acts as a data exchange platform between aircraft manufacturers, airlines, leasing companies, and MRO. In the present example, centralized ledger **302** provides a Packaged Business Capability (PBC) that might handle several services or functions related to aircraft maintenance such as work orders, parts availability, aircraft transfer/return, etc.

(34) In contrast to FIG. 2, wherein each party/platform has direct point-to-point interface with every other, centralized ledger **302** requires only a single interface each for subscribers such as maintenance applications **304**, airlines **306**, MROs **308**, lessors **310**, etc. Centralized ledger **302** eliminates the need for custom scripts and interfaces and supports industry standards for aircraft and maintenance data.

(35) FIG. 4 depicts a diagram of an architecture **400** for a centralized ledger maintenance system in accordance with an illustrative embodiment.

(36) A ledger dashboard **404** provides status updates to users **440**. The ledger dashboard **404** interacts with the maintenance ledger **406** via a ledger API **408**. Ledger API **408** allows users **440** to review all ledger items according to access permissions. Ledger dashboard **404** may have an authorized interface to support enhanced smart contracts.

(37) The maintenance ledger **406** is immutable and comprises smart workflows and notifications. The centralized maintenance ledger **406** includes a ledger database **410** which provides cryptographic verifiability and a complete history of all transactions. Ledger database **410** might be, for example, a structured query language (SQL) database.

(38) The maintenance ledger **406** also includes a notification engine **412** and may also include a business rules engine **414** and workflow engine **416**.

(39) The centralized maintenance ledger **406** operates in a ledger capability system **402** that can be deployed in a cloud environment **420**. Customized platforms **430**, **432** for airlines and MROs can also interact with the centralized maintenance ledger **406** through API gateway **422**. Customized platforms **430**, **432** may also be part of ledger dashboard **404**.

(40) FIG. 5 illustrates the operation of the centralized ledger through the use of enhanced smart contracts in accordance with an illustrative embodiment. The centralized ledger **502** provides a single view of truth for all participating subscribing partners **504**, **506**. The enhanced smart contract comprises a set of rules that is stored on the ledger blockchain of centralized ledger **502** and is automatically executed. For example, the enhanced smart contract sends automatic notification of new activity. As another example, the enhanced smart contract can monitor compliance activities of business process model fragments and provide notice when compliance is achieved.

(41) The centralized ledger **502** manages maintenance with regulatory requirements. The enhanced smart contract controls the workflow automatically and provides notifications to all subscribing partners **504**, **506**. Each subscribing partner **504**, **506** uses the same centralized ledger **502** for material management, maintenance control, and maintenance planning. The centralized ledger **502** generates a resource planning “twin” **508** of the subscribing partners **504**, **506** which provides a paperless log book, health monitoring, document management, and material management.

(42) The distributed centralized ledger **502** allows all subscribing partners **504**, **506** to have access to the centralized ledger and its immutable record of transactions. Due to the immutable record of transaction no partner can change a transaction after it is recorded in the blockchain of the centralized ledger **502**. To correct an error, a new transaction is required to “reverse” or negate the error in question. Only specified partners have authority for a specific transaction (e.g., changing the current activity state) according to prescribed access permissions.

(43) FIG. 6 illustrates examples of types of entries into the centralized ledger. FIGS. 7A, 7B, and 7C illustrate an example process flow for a work order transaction based on the types of entries shown in FIG. 6.

(44) Process flow **700** begins with a controller assessment event **702** resulting in the manual initiation **704** of a work order **602** based on the assessment. In the present example, the new work order is initiated by a health maintenance service (HMS) which is a partner/subscriber to the centralized ledger. When the new work order is entered into the centralized ledger, an enhanced smart contract (compliance contract) is executed, and a notification is issued indicating that the work order has been initiated.

(45) A content management service (CMS) partner that is subscribed to new ledger notifications receives notice of the new work order and adds **706** required actions **604** related to the work order to the centralized ledger. In response, the enhanced smart contract executes and another notification that the maintenance actions have been added to the centralized ledger is issued.

(46) A maintenance planner partner subscribed to task allocation notifications receives the notification of the require actions **604** added to the centralized ledger. The maintenance planning partner inputs **708** scheduling details **606** into the centralized ledger. These details may comprise planning schedules, skills, and other details required to complete the work order in question. The

compliance contract executes and verifies that all contractual and regulatory requirements have been met to execute the work order. Another notification is raised indicating that a maintenance work order is now ready to be executed.

(47) A maintenance execution partner subscribed to notifications for work orders ready for execution receives the notification from the centralized ledger that the work order is ready for execution according to the scheduling details **606**. The maintenance execution partner enters the tasks performed and disposition into the centralized ledger as maintenance tasks are completed. Upon completion, a technician digitally signs to verify action completion **608** of the tasks per contractual compliance.

(48) The compliance contract executes to verify that all contract objects have been met **712**. Upon verification of completion of work tasks **610** according to contractual requirements a final check of the compliance contract is made, and the work order is closed.

(49) If any of the required actions is determined not to be in compliance with regulations, the action is placed in a deferred stated, and a deferred work order **612** is generated to be executed in the future.

(50) The centralized ledger therefore provides a single view of truth with transparency across business partners. The immutable ledger captures partner compliance action. The centralized ledger automatically validates compliance and execution of processes. The centralized ledger also eliminated the need for maintaining point-to-point partner integrations. Compliance contracts can be supported as enhanced smart contracts assigned as ledger entries.

(51) FIG. **8** depicts a flowchart of a process for processing a vehicle maintenance transaction in accordance with an illustrative embodiment. Process **800** can be implemented in hardware, software, or both. When implemented in software, the process can take the form of program code that is run by one of more processor units located in one or more hardware devices in one or more computer systems. For example, process **800** can be implemented in maintenance record system **100** in FIG. **1**.

(52) Process **800** begins by receiving the transaction (operation **802**) and entering the transaction into a blockchain ledger (operation **804**).

(53) In response to the transaction, an enhanced smart contract in the blockchain ledger executes (operation **806**). The enhanced smart contract specifies a number of compliance requirements to complete the transaction and automates a workflow for the transaction according to the compliance requirements. The compliance requirements might comprise a number of tasks required for the transaction, technical knowledge required by the tasks, and scheduling of the tasks. The enhanced smart contract might retrieve the compliance requirements for the transaction from a number of data sources.

(54) Process **800** generates a number of notifications regarding the transaction to a number of subscribers to the blockchain ledger (operation **808**). The subscribers might include manufacturer applications, airlines, and maintenance and repair organizations. The enhanced smart contract may limit access of the subscribers to records in the blockchain ledger according to access permissions. The enhanced smart contract may control which subscribers receive notifications according to the access permissions.

(55) The blockchain ledger receives from the subscribers confirmation of fulfillment of the compliance requirements to complete the transaction (operation **810**).

(56) The enhanced smart contract verifies fulfillment of the compliance requirements (operation **812**). Upon completion of the transaction, the completion is recorded in the blockchain ledger. Process **800** then ends. The record in the blockchain ledger of completion of the transaction can only be negated by creating a new transaction.

(57) The flowchart and block diagrams in the different depicted embodiments illustrate the architecture, functionality, and operation of some possible implementations of apparatuses and methods in an illustrative embodiment. In this regard, each block in the flowcharts or block



diagrams can represent at least one of a module, a segment, a function, or a portion of an operation or step. For example, one or more of the blocks can be implemented as program code, hardware, or a combination of the program code and hardware. When implemented in hardware, the hardware can, for example, take the form of integrated circuits that are manufactured or configured to perform one or more operations in the flowcharts or block diagrams. When implemented as a combination of program code and hardware, the implementation may take the form of firmware. Each block in the flowcharts or the block diagrams can be implemented using special purpose hardware systems that perform the different operations or combinations of special purpose hardware and program code run by the special purpose hardware.

(58) In some alternative implementations of an illustrative embodiment, the function or functions noted in the blocks may occur out of the order noted in the figures. For example, in some cases, two blocks shown in succession may be performed substantially concurrently, or the blocks may sometimes be performed in the reverse order, depending upon the functionality involved. Also, other blocks may be added in addition to the illustrated blocks in a flowchart or block diagram.

(59) Turning now to FIG. 9, an illustration of a block diagram of a data processing system is depicted in accordance with an illustrative embodiment. Data processing system **900** may be used to implement computer system **150** in FIG. 1. In this illustrative example, data processing system **900** includes communications framework **902**, which provides communications between processor unit **904**, memory **906**, persistent storage **908**, communications unit **910**, input/output (I/O) unit **912**, and display **914**. In this example, communications framework **902** takes the form of a bus system.

(60) Processor unit **904** serves to execute instructions for software that may be loaded into memory **906**. Processor unit **904** may be a number of processors, a multi-processor core, or some other type of processor, depending on the particular implementation. In an embodiment, processor unit **904** comprises one or more conventional general-purpose central processing units (CPUs). In an alternate embodiment, processor unit **904** comprises one or more graphical processing units (GPUS).

(61) Memory **906** and persistent storage **908** are examples of storage devices **916**. A storage device is any piece of hardware that is capable of storing information, such as, for example, without limitation, at least one of data, program code in functional form, or other suitable information either on a temporary basis, a permanent basis, or both on a temporary basis and a permanent basis. Storage devices **916** may also be referred to as computer-readable storage devices in these illustrative examples. Memory **906**, in these examples, may be, for example, a random access memory or any other suitable volatile or non-volatile storage device. Persistent storage **908** may take various forms, depending on the particular implementation.

(62) For example, persistent storage **908** may contain one or more components or devices. For example, persistent storage **908** may be a hard drive, a flash memory, a rewritable optical disk, a rewritable magnetic tape, or some combination of the above. The media used by persistent storage **908** also may be removable. For example, a removable hard drive may be used for persistent storage **908**. Communications unit **910**, in these illustrative examples, provides for communications with other data processing systems or devices. In these illustrative examples, communications unit **910** is a network interface card.

(63) Input/output unit **912** allows for input and output of data with other devices that may be connected to data processing system **900**. For example, input/output unit **912** may provide a connection for user input through at least one of a keyboard, a mouse, or some other suitable input device. Further, input/output unit **912** may send output to a printer. Display **914** provides a mechanism to display information to a user.

(64) Instructions for at least one of the operating system, applications, or programs may be located in storage devices **916**, which are in communication with processor unit **904** through communications framework **902**. The processes of the different embodiments may be performed by

processor unit **904** using computer-implemented instructions, which may be located in a memory, such as memory **906**.

(65) These instructions are referred to as program code, computer-usable program code, or computer-readable program code that may be read and executed by a processor in processor unit **904**. The program code in the different embodiments may be embodied on different physical or computer-readable storage media, such as memory **906** or persistent storage **908**.

(66) Program code **918** is located in a functional form on computer-readable media **920** that is selectively removable and may be loaded onto or transferred to data processing system **900** for execution by processor unit **904**. Program code **918** and computer-readable media **920** form computer program product **922** in these illustrative examples. In one example, computer-readable media **920** may be computer-readable storage media **924** or computer-readable signal media **926**.

(67) In these illustrative examples, computer-readable storage media **924** is a physical or tangible storage device used to store program code **918** rather than a medium that propagates or transmits program code **918**. Computer readable storage media **924**, as used herein, is not to be construed as being transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide or other transmission media (e.g., light pulses passing through a fiber-optic cable), or electrical signals transmitted through a wire, as used herein, is not to be construed as being transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide or other transmission media (e.g., light pulses passing through a fiber-optic cable), or electrical signals transmitted through a wire.

(68) Alternatively, program code **918** may be transferred to data processing system **900** using computer-readable signal media **926**. Computer-readable signal media **926** may be, for example, a propagated data signal containing program code **918**. For example, computer-readable signal media **926** may be at least one of an electromagnetic signal, an optical signal, or any other suitable type of signal. These signals may be transmitted over at least one of communications links, such as wireless communications links, optical fiber cable, coaxial cable, a wire, or any other suitable type of communications link.

(69) The different components illustrated for data processing system **900** are not meant to provide architectural limitations to the manner in which different embodiments may be implemented. The different illustrative embodiments may be implemented in a data processing system including components in addition to or in place of those illustrated for data processing system **900**. Other components shown in FIG. **9** can be varied from the illustrative examples shown. The different embodiments may be implemented using any hardware device or system capable of running program code **918**.

(70) As used herein, the phrase “at least one of,” when used with a list of items, means different combinations of one or more of the listed items can be used, and only one of each item in the list may be needed. In other words, “at least one of” means any combination of items and number of items may be used from the list, but not all of the items in the list are required. The item can be a particular object, a thing, or a category.

(71) For example, without limitation, “at least one of item A, item B, or item C” may include item A, item A and item B, or item B. This example also may include item A, item B, and item C or item B and item C. Of course, any combinations of these items can be present. In some illustrative examples, “at least one of” can be, for example, without limitation, two of item A; one of item B; and ten of item C; four of item B and seven of item C; or other suitable combinations.

(72) As used herein, “a number of” when used with reference to items, means one or more items. For example, “a number of different types of networks” is one or more different types of networks. In illustrative example, a “set of” as used with reference items means one or more items. For example, a set of metrics is one or more of the metrics.

(73) The description of the different illustrative embodiments has been presented for purposes of

illustration and description and is not intended to be exhaustive or limited to the embodiments in the form disclosed. The different illustrative examples describe components that perform actions or operations. In an illustrative embodiment, a component can be configured to perform the action or operation described. For example, the component can have a configuration or design for a structure that provides the component an ability to perform the action or operation that is described in the illustrative examples as being performed by the component. Further, to the extent that terms “includes”, “including”, “has”, “contains”, and variants thereof are used herein, such terms are intended to be inclusive in a manner similar to the term “comprises” as an open transition word without precluding any additional or other elements.

(74) Many modifications and variations will be apparent to those of ordinary skill in the art. Further, different illustrative embodiments may provide different features as compared to other desirable embodiments. The embodiment or embodiments selected are chosen and described in order to best explain the principles of the embodiments, the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

## Claims

1. A computer-implemented method for transmitting notifications of maintenance work on a vehicle, the method comprising using a number of processors to perform: receiving transactions from multiple sources with different data integration patterns and models through a single application programming interface, the multiple sources comprising applications comprising at least: health monitoring, document management, mobile documents, a paperless logbook, structures, a parts catalog, directives, and records, for the vehicle; the single application programming interface transforming each of the different data integration patterns and models for entering the transactions into a blockchain ledger serving as a single point of truth for all transactions regarding the vehicle; in response to the transactions, executing an enhanced smart contract in the blockchain ledger, wherein the enhanced smart contract specifies a number of compliance requirements to complete the transactions and creates a workflow for the transactions that will fulfill all the compliance requirements; generating, in a notification engine, and transmitting a number of notifications regarding the transactions to a number of subscribers to the blockchain ledger comprising at least: airlines, maintenance and repair organizations, and aircraft lessors, and thereby eliminating a need for custom scripts and point-to-point interfacing between the number of subscribers; receiving, from the subscribers, confirmation of fulfillment of the compliance requirements to complete the transactions; verifying, by the enhanced smart contract, fulfillment of the compliance requirements; and upon completion of the transactions and fulfillment of the compliance requirements, recording the completion in the blockchain ledger.
2. The method of claim 1, wherein the compliance requirements comprise: a number of tasks required for the transactions; technical knowledge required by the tasks; and scheduling of the tasks.
3. The method of claim 1, wherein the enhanced smart contract retrieves the compliance requirements for the transactions from the multiple sources.
4. The method of claim 1, wherein a record in the blockchain ledger can only be negated by creating a new transaction.
5. The method of claim 1, wherein the enhanced smart contract limits access of the subscribers to records in the blockchain ledger according to access permissions.
6. The method of claim 5, wherein the enhanced smart contract controls which subscribers receive notifications according to the access permissions.
7. The method of claim 1, wherein the subscribers to the blockchain ledger include at least one of: maintenance applications; airlines; or maintenance and repair organizations.

8. The method of claim 1, wherein the blockchain ledger generates a resource planning twin of the subscribers.

9. A system configured to transmit notifications for maintenance work on a vehicle, the system comprising: a storage device that stores program instructions; one or more processors operably connected to the storage device and configured to execute the program instructions to cause the system to: receive transactions from multiple sources with different data integration patterns and models through a single application programming interface, the multiple sources comprising applications comprising at least: health monitoring, document management, mobile documents, a paperless logbook, structures, a parts catalog, directives, and records, for the vehicle; transform each of the different data integration patterns and models and enter the transactions into a blockchain ledger that creates a single point of truth for all transactions about the vehicle; in response to the transactions, execute an enhanced smart contract in the blockchain ledger, wherein the enhanced smart contract specifies a number of compliance requirements to complete the transactions and creates a workflow for the transactions that fulfills all the compliance requirements; generate and transmit through a notification engine a number of notifications regarding the transactions to a number of subscribers to the blockchain ledger that comprise at least: airlines, maintenance and repair organizations, and aircraft lessors, wherein the number of notifications eliminate a need for custom scripts and point-to-point interfaces among the number of subscribers; receive, from the subscribers, confirmation of fulfillment of the compliance requirements to complete the transactions; verify, by the enhanced smart contract, fulfillment of the compliance requirements; and upon completion of the transactions and fulfillment of the compliance requirements, record the completion in the blockchain ledger.

10. The system of claim 9, wherein the compliance requirements comprise: a number of tasks required for the transactions; technical knowledge required by the tasks; and scheduling of the tasks.

11. The system of claim 9, wherein the enhanced smart contract retrieves the compliance requirements for the transactions from the multiple sources.

12. The system of claim 9, wherein a record in the blockchain ledger can only be negated by creating a new transaction.

13. The system of claim 9, wherein the enhanced smart contract limits access of the subscribers to records in the blockchain ledger according to access permissions.

14. The system of claim 13, wherein the enhanced smart contract controls which subscribers receive notifications according to the access permissions.

15. The system of claim 9, wherein the subscribers to the blockchain ledger include at least one of: maintenance applications; airlines; or maintenance and repair organizations.

16. The system of claim 9, wherein the blockchain ledger generates a resource planning twin of the subscribers.

17. A computer program product configured to transmit notifications of maintenance requirements of a vehicle, wherein the computer program product comprises: a computer-readable storage medium having program instructions embodied thereon to perform the steps of: receiving transactions from multiple sources with different data integration patterns and models through a single application programming interface, the multiple sources comprising applications comprising at least: health monitoring, document management, mobile documents, a paperless logbook, structures, a parts catalog, directives, and records, for the vehicle; the single application programming interface transforming each of the different data integration patterns and models for entering the transactions into a blockchain ledger serving as a single point of truth for all transactions regarding the vehicle; in response to the transactions, executing an enhanced smart contract in the blockchain ledger, wherein the enhanced smart contract specifies a number of compliance requirements to complete the transactions and creates a workflow for the transactions that will fulfill all the compliance requirements; generating, in a notification engine, and

transmitting a number of notifications regarding the transactions to a number of subscribers to the blockchain ledger comprising at least: airlines, maintenance and repair organizations, and aircraft lessors, and thereby eliminating a need for custom scripts and point-to-point interfacing between the number of subscribers; receiving, from the subscribers, confirmation of fulfillment of the compliance requirements to complete the transactions; verifying, by the enhanced smart contract, fulfillment of the compliance requirements; and upon completion of the transactions and fulfillment of the compliance requirements, recording the completion in the blockchain ledger.

18. The computer program product of claim 17, wherein the compliance requirements comprise: a number of tasks required for the transactions; technical knowledge required by the tasks; and scheduling of the tasks.

19. The computer program product of claim 17, wherein the enhanced smart contract retrieves the compliance requirements for the transactions from a number of authorized data sources.

20. The computer program product of claim 17, wherein a record in the blockchain ledger can only be negated by creating a new transaction.

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